STUDENT TREND ANALYSIS

Case Study Report

This Python program compares students' marks in two examinations across three subjects. It identifies improvements and declines, checks for weak subjects, calculates totals and the second-exam average, and assigns a grade based on the average.

1. Problem Statement

Develop a Python program to analyze a student's performance in two examinations for Maths, Python, and English. The program should compare marks subject by subject, report whether each subject improved or declined, identify subjects with low marks in both examinations, calculate the exam totals and average, and display the student's grade.

2. Objective

The objective is to analyze changes in academic performance using Python lists, loops, conditional statements, arithmetic operations, and variables. The program helps summarize marks and identify areas where a student may need improvement.

3. Assumptions

• The subjects are Maths, Python, and English.

• Exam 1 marks are 40, 60, and 50 respectively.

• Exam 2 marks are 55, 70, and 45 respectively.

• Each subject is treated equally when calculating the average.

• A subject is considered weak if the marks in both exams are below 40.

• An average of 75 or above is Excellent; 50 or above but below 75 is Average; below 50 Needs Improvement.

4. Algorithm

1. Start the program.

2. Create a list of subjects and lists containing Exam 1 and Exam 2 marks.

3. Initialize both exam totals to zero.

4. Loop through each subject.

5. Calculate the difference between Exam 2 and Exam 1 marks.

6. Print whether the marks improved, declined, or stayed the same.

7. If both marks are below 40, print a weak-subject warning.

8. Add the marks to the respective exam totals.

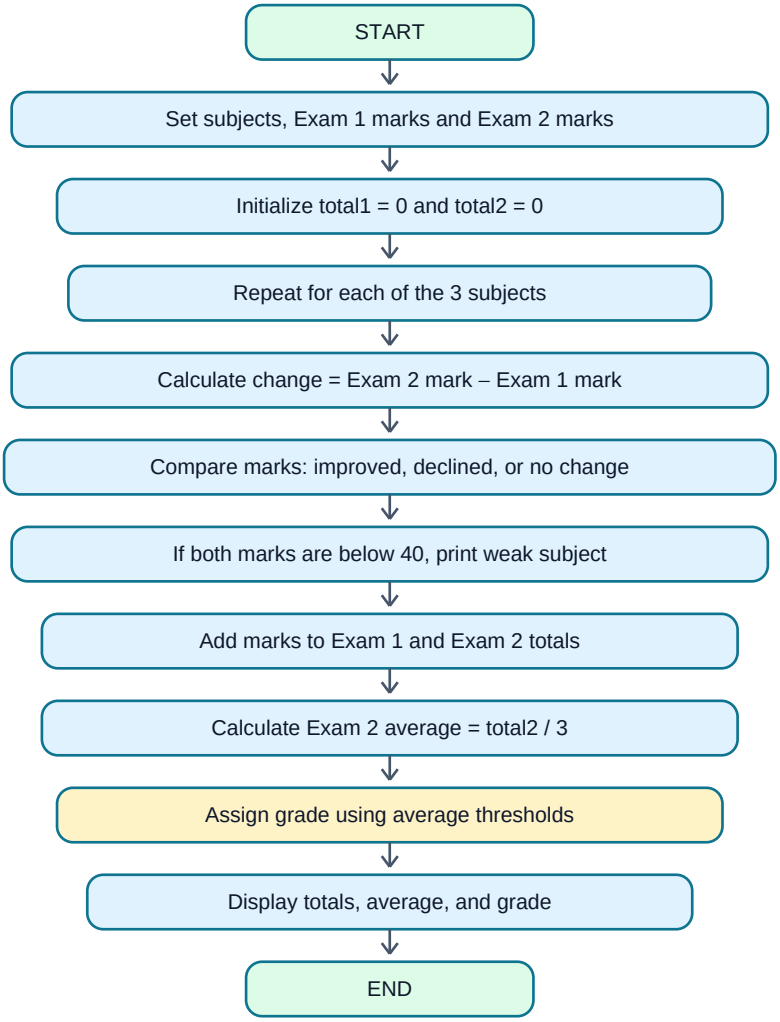
9. Display the totals and calculate the Exam 2 average.

10. Assign a grade based on the average and display it.

11. Stop the program.

5. Flowchart

The flowchart presents the main steps followed by the program, from reading the marks to comparing subject performance and displaying the final grade.



6. Pseudocode

```
BEGIN
    subjects ← ["Maths", "Python", "English"]
    exam1 ← [40, 60, 50]
    exam2 ← [55, 70, 45]
    total1 ← 0
    total2 ← 0

    FOR i ← 0 TO 2
        PRINT subjects[i]
        change ← exam2[i] – exam1[i]

        IF change > 0
            PRINT "Improved by", change
        ELSE IF change < 0
            PRINT "Declined by", -change
        ELSE
            PRINT "No change"
        END IF

        IF exam1[i] < 40 AND exam2[i] < 40
            PRINT "Weak subject"
        END IF

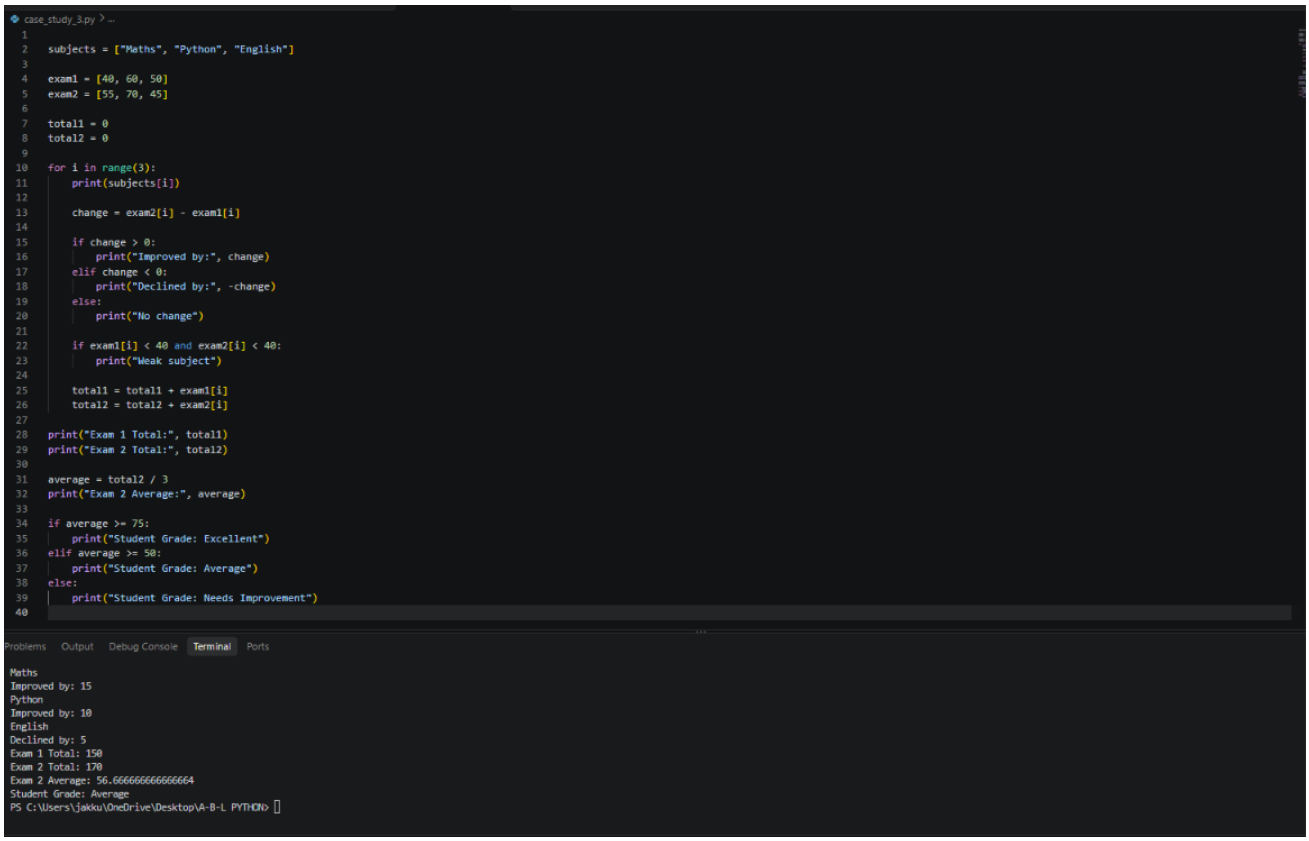
        total1 ← total1 + exam1[i]
        total2 ← total2 + exam2[i]
    END FOR

    PRINT total1, total2
    average ← total2 / 3
    PRINT average

    IF average ≥ 75
        PRINT "Excellent"
    ELSE IF average ≥ 50
        PRINT "Average"
    ELSE
        PRINT "Needs Improvement"
    END IF
END
```

7. Screenshot of Python Source Code and Output

The original screenshot of the Python source code and its terminal output is included below.



8. Conclusion

The Student Trend Analysis program compares marks across two examinations and summarizes the student's performance. For the given data, Maths improves by 15 marks, Python improves by 10 marks, and English declines by 5 marks. Exam 1 total is 150, Exam 2 total is 170, and the Exam 2 average is approximately 56.67, so the program assigns the grade Average. This case study demonstrates how Python can be used to process marks and present useful academic performance information.